STOCHASTIC METHODS IN FINANCE
STAT0013
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Suggested Solutions to Exercises 5

1. The stock price S_t follows geometric Brownian motion, so the SDE for the stock is
$$dS_t = \mu S_t dt + \sigma S_t dB_t,$$
where B_t is standard Brownian motion. To obtain the SDE for S_t^2 we use Itô's lemma for the function $G_t = G(S_t, t) = S_t^2$. We have
$$\frac{\partial G}{\partial S} = 2S; \qquad \frac{\partial^2 G}{\partial S^2} = 2; \qquad \frac{\partial G}{\partial t} = 0.$$
Itô's lemma then gives us
$$\begin{aligned} dG_t &= 2S_t\, dS_t + \tfrac{1}{2}2\,(dS_t)^2 \\ &= \big(2\mu S_t^2 + \sigma^2 S_t^2\big)dt + 2\sigma S_t^2 dB_t \\ &= (2\mu + \sigma^2)S_t^2 dt + 2\sigma S_t^2 dB_t \\ &= (2\mu + \sigma^2)G_t dt + 2\sigma G_t dB_t \end{aligned}$$
Thus, G_t – the square of the stock price – follows another geometric Brownian motion, with drift parameter $2\mu+\sigma^2$ and volatility parameter 2σ.

2. (a) Write the ODE as
$$\begin{aligned} & dx(t) = x(t)\,(a + bg'(t))\,dt; \\ \Rightarrow \qquad & \frac{dx(t)}{x(t)} = (a + bg'(t))\,dt. \end{aligned}$$
Then, taking integrals on both sides, we have
$$\log x(t) + C = at + bg(t),$$

where C is the constant of integration. Using the two given initial conditions, we find that $C = -\log x_0$ which gives

$$x(t) = x_0 e^{at+bg(t)}$$

as required.

(b) We have that $X_t = x\, e^{at+bW_t} = X(t, W_t)$. Notice that

$$\frac{\partial X}{\partial W} = xbe^{at+bW_t} = bX_t;$$
$$\frac{\partial^2 X}{\partial W^2} = xb^2 e^{at+bW_t} = b^2 X_t;$$
$$\frac{\partial X}{\partial t} = axe^{at+bW} = aX_t.$$

Thus, using Itô's lemma on X_t, gives that

$$dX_t = \big(aX_t + \tfrac{b^2}{2} X_t\big)dt + bX_t dW_t,$$

as required. For the initial condition, $X_0 = xe^{bW_0}$, and by the properties of Brownian motion we know that $W_0 = 0$, so that $X_0 = x$.

Note the extra term $\frac{b^2}{2} X_t$ when moving from the ODE world in (2a) (where $g(t)$ was a deterministic function) to the SDE world in part (2b) (where W_t is stochastic Brownian motion).

3. We write $G(t, x) = e^x e^{-t/2}$ and then apply Itô's lemma to this function. The required partial derivatives are

$$\frac{\partial G}{\partial t} = -\tfrac{1}{2} e^x e^{t/2}; \qquad \frac{\partial G}{\partial x} = e^x e^{t/2}; \qquad \frac{\partial^2 G}{\partial x^2} = e^x e^{t/2}.$$

We have $X_t = e^{W_t} e^{-t/2}$, so using Itô's lemma we get

$$\begin{aligned} dX_t &= \big(\tfrac{\partial G}{\partial t}(t, W_t) + \tfrac{1}{2}\tfrac{\partial^2 G}{\partial x^2}(t, W_t)\big)dt + \tfrac{\partial G}{\partial x}(t, W_t)dW_t \\ &= (-\tfrac{1}{2} X_t + \tfrac{1}{2} X_t)dt + X_t dW_t \\ &= X_t dW_t \end{aligned}$$

as required.

4. The price of the zero-coupon government bond is the present value of a guaranteed future payment of £100, which is

$$B_t = 100\, e^{-\int_t^T r_s ds}.$$

We now apply Itô's formula on $B_t = 100\, e^{X_t}$, where $X_t = -\int_t^T r_s ds$. Notice that $dX_t = r_t dt$. Thus, we obtain,

$$\begin{aligned} dB_t &= 100\, e^{X_t} dX_t + \tfrac{1}{2} 100\, e^{X_t} (dX_t)^2 \\ &= 100\, e^{X_t} dX_t + \tfrac{1}{2} 100\, e^{X_t} (r_t)^2 (dt)^2 \\ &= 100\, e^{X_t} dX_t + 0 \\ &= B_t\, r_t\, dt \end{aligned}$$

The initial condition is $B_0 = 100\, e^{X_0} = 100\, e^{-\int_0^T r_s ds}$. In constrast with all other cases we have considered, this is not a deterministic initial condition.